MCQ 1: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $e^0 =$
   - A. 0
   - B. 1
   - C. e
   - D. -1
2. $e^x e^y =$
   - A. e^{x+y}
   - B. e^{xy}
   - C. $e^x + e^y$
   - D. e^{x-y}
3. Derivative of e^{-2x}:
   - A. e^{-2x}
   - B. $2e^{-2x}$
   - C. $-e^{-2x}$
   - D. $-2e^{-2x}$
4. $e^x = 1$ is equivalent to
   - A. $x = 1$
   - B. $x = e$
   - C. $x = 0$
   - D. no solution
5. The sign of e^{-x} is
   - A. negative if x>0
   - B. always strictly positive
   - C. zero at 0
   - D. variable

MCQ 2: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $e^{2x} < e^3$ is equivalent to
   - A. $x < 3$
   - B. $x > \frac{3}{2}$
   - C. $x < \frac{3}{2}$
   - D. $x < \sqrt{3}$
2. $u_n = e^{3n}$ has common ratio
   - A. e^3

Booklet page 82

B. 3
C. e
D. $3e$

3. Tangent to e^x at 0:
   A. $y = x$
   B. $y = ex$
   C. $y = 1$
   D. $y = x + 1$
4. Derivative of xe^x:
   A. e^x
   B. $(x+1)e^x$
   C. xe^x
   D. xe^{x-1}
5. $e^x \geq 1 + x$ has equality
   A. for all x
   B. at 1 only
   C. at 0 only
   D. never

True or false? Justify.

1. $e^{2x} = 2e^x$ for all x.
2. The function $t \mapsto e^{-3t}$ is decreasing.
3. A continuous rate k=0.1 corresponds exactly to a 10 % increase in one unit of time.

> **Key point · Self-assessment.** I keep the rules for powers; I know the positivity; I differentiate with the factor; I separate model, exact calculation and simulation.
>
> For each objective, indicate: achieved without help; achieved with help; to be reworked. Write one example and one mistake to avoid.